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Capstone - Midterm Report:

SUSHI ASSEMBLY LINE

PROCTECH 4TR1– C03

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Project Update



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PROCTECH 4TR1: Capstone Design Project I

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Contents

Abstract 4

Introduction 5

Project Objectives 6

Project Background & Literature Review 7

Design Concept & Selected Solution 9

Discussion 13

Conclusion 15

References 16

Abstract

Our capstone project is a sushi roll production machine. The general problem we wish to solve is that the standard sushi roll, although very popular around the world and relatively simple in construction, takes time to make. Our device aims to eliminate the time required to prepare these simpler rolls.

The basic assembly of a sushi roll can be broken down into five simple steps: (1) lay out an even sheet of rice and nori; (2) add the roll's ingredients in a line across the sheet; (3) roll the whole assembly into an even cylinder; (4) slice the roll into evenly spaced pieces; and (5) plate the pieces. Our device will have a designated station and mechanism for each of these steps. It will also feature a platform that moves from side to side to safely and precisely move the roll through the machine's processes. The design will also emphasize safety features to prevent harm to the user and will be completely food safe.

This proposed solution supports chefs by automating repetitive preparation steps, freeing them to focus on higher-value culinary tasks. If successfully implemented, this system could serve both as an operational aid in restaurants and as a prototype for larger-scale food manufacturing applications.

Introduction

Sushi is a beloved food all around the world. Sushi restaurants and grocery retailers face a consistent operational challenge: basic sushi rolls are in extremely high demand, yet they require significant time and repetitive manual effort to produce. Although these rolls are simple in design, the preparation process involves multiple sequential steps that must be performed carefully and consistently. This creates a bottleneck in both restaurant kitchens and large-scale grocery production settings.

The problem manifests most clearly during peak service hours and high-volume production periods. Orders accumulate, preparation times increase, and skilled chefs must devote large portions of their time to repetitive tasks rather than higher-value culinary work. In grocery supply chains, slow production limits output capacity and reduces the ability to meet growing consumer demand for ready-to-eat sushi products.

This inefficiency leads to several difficulties. Restaurants may experience longer wait times and operational strain, while grocery producers face constraints in scaling output. Labour costs increase when highly trained chefs spend time on routine assembly rather than specialized dishes. Over time, this imbalance reduces productivity, increases pressure on staff, and limits overall growth potential.

At the beginning of this project, our team expected to explore whether automation could meaningfully reduce these inefficiencies without compromising quality. We aimed to design a solution that would streamline production, improve consistency, and support both small-scale restaurant operations and larger industrial environments.

To address this need, our project proposes a sushi assembly line concept that automates repetitive preparation tasks while maintaining product quality and consistency. At a high level, the system is designed to support chefs rather than replace them, freeing their time for complex creations while enabling scalable production for grocery distribution. The goal is to increase efficiency, reduce strain on skilled labour, and create a more sustainable production model for high-demand sushi products. The system will also be a prototype for a large-scale assembly line for food plants.

Project Objectives

The proposed objectives for this automated sushi assembly line are as follows:

- It shall produce uniform pieces of sushi that are consistent in appearance and quality
- It shall not require human intervention to complete the process, aside from replenishing the ingredients/materials
- It shall be safe for humans to interact with the system if need be
- It shall be entirely food-grade and up to food safety requirements
- It shall have an HMI to display relevant data and allow for emergency control (ex. runtime data, emergency stops, etc.)
- It shall be quick and efficient, taking less time than it would for an average human to execute the process

Project Background & Literature Review

Automation of sushi preparation is not a brand-new concept. There are existing commercial machines designed to perform specific sushi-making tasks that have been adopted in some restaurants and food service environments. A notable example in this field is the lineup of commercial sushi machines from AUTEC.



A rice roller machine from AUTEC.

AUTEC is a company with a long history in sushi automation tools dating back to the 1980s when they released some of the first commercial sushi forming devices and have continued expanding their product range globally [1].

The issue with these machines is that they are focused on individual production steps such as creating rice sheets or cutting rolls. This is the current state of sushi machines, or the "prior work" we are building on.

The gap this project seeks to build on is to move beyond isolated machines that automate *parts* of sushi preparation toward a more integrated assembly line approach. Instead of simply producing rice balls or singular rolls, the proposed system aims to connect and coordinate tasks into a streamlined workflow, reducing the manual intervention still present in existing machines and supporting high-volume production for markets like restaurants and grocery distribution. Our proposed solution will include a full assembly line for a sushi roll from scratch,

with multiple substations on a small table with wheels for placing ingredients, rolling the sushi, cutting, and placing on a plate, ready to serve.

Design Concept & Selected Solution

Like any good assembly line, this project can be broken up into the following workstations:

- *Station 0:* Linear Slide
- *Station 1:* Flattening the Rice
- *Station 2:* Dispensing the Neta
- *Station 3:* Rolling the Sushi
- *Station 4:* Cutting the Roll
- *Station 5:* Plating the Pieces

Each station is being designed independently, with the outputs of one station being expected to become the inputs of the subsequent station. This encourages a seamless integration of workstations when all is put together.

Function	Option 1	Option 2	Option 3
Foundation	Conveyor Belt (Continuous Process)	Linear Guide Rail (Batch Process)	
Preliminary Set-Up	Place Nori in Center of Bamboo Mat; Select Type of Sushi Roll; Place Plate at End; Press Start;	Place Nori with Rice in Center of Bamboo Mat; Select Type of Sushi Roll; Place Plate at End; Press Start	Place Nori in Center of Bamboo Mat; Place Plate at End; Press Start (No Sushi Type Selection)
Station 1: Flattening the Rice	Hopper of Rice; 4 Wooden Rollers; Machine Moves Over Platform	Hopper of Rice; 2 Silicon Rollers; Platform Moves Under Machine	
Station 2: Dispensing the Neta	Single Column with Individual Compartments; Platform Moves Under Machine; Compartments Rotate to Drop Ingredients One By One	Single Wide Column with Individual Compartments; Platform Stops Under Machine; Floor of Each Compartment Opens to Drop All Ingredients At Once	Two Single Wide Columns with Individual Compartments - One for Fish, One for Avocado; Platform Stops Under Machine; Floor of Each Compartment Opens to Drop All Ingredients At Once
Station 3: Rolling the Sushi	Prongs That Protrude From Moving Platform; Grips Bamboo Mat and Roll From Bottom; Tightens Using Stepper Motor	Stationary Prongs; Platform Moves Under Machine; Scoops Bamboo Mat and Roll From Above; Tightens Using Stepper Motor	Stationary Machine Attaches to One Edge of Bamboo Mat; Folds Over the Roll; Tightens Using Stepper Motor
Station 4: Cutting the Roll	Series of 5 Parallel Straight Blades; Machine Comes Down Onto Bamboo Mat and Presses; Plastic Guards Surround Blades and Retract When Pressed	Series of 5 Parallel Straight Blades; Machine Comes Down In Front of Bamboo Mat and Moves Back to Cut Roll; Large Plastic Guards Encapsulate Roll and Blades	Series of 5 Parallel Seraded Blades; Machine Performs Sawing Motion to Cut Roll; Large Plastic Guards Encapsulate Roll and Blades
Station 5: Plating the Pieces	Pick-and-Place End Effector; Camera Vision to Locate Sushi Pieces	Pick-and-Place End Effector; Hardcoded Pick-Up and Drop-Off Locations	

Morphological chart with options for each assembly line substation.

Regarding the foundation of the system, our group had to decide between using a conveyor belt or a linear guide rail to transport the unfinished roll from station to station. The former lends itself to continuous processes due to being able to have multiple rolls placed on it, while the latter suggests a continuous process as only a single roll will be carried from end-to-end. Our group ultimately decided on using a linear guide rail system as this would emphasize quality and consistency in our rolls, over efficiency. Having a single platform that carries the roll from station to station also allows for more flexibility, and the guide rails allow for easy installation of limit switches which will tell the microcontrollers exactly where the platform is located.

Before the automated line begins, certain tasks need to be completed by the operator/chef. With the moving platform at the start, a bamboo mat should be placed with a sheet of nori

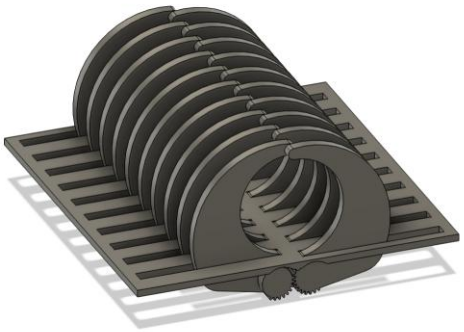
centered on it. A plate should also be placed at the end of the assembly line. Finally, the operator should select the desired nata (fillings) for the sushi roll and then press START on the HMI. Occasionally, the operator will need to replenish the ingredients in the compartments in Station 2 as ingredients like fish and avocado should not be left out at room temperature for longer than two hours [2].

With the assembly line running, the platform stops at Station 1: Flattening the Rice. The goal of this station is to dispense a bed of rice neatly onto the sheet of nori. This machine has a hopper of cooked sushi rice at the top, which is fed into a pair of silicon rollers that flatten the rice to a certain thickness. We decided that having the platform and sheet of nori slowly move forward under the machine would be the best way to achieve a uniform bed of rice, as opposed to having the machine move over a stationary platform.

Next, the platform will stop at Station 2: Dispensing the Neta. This station intends to lay out the ingredients (fish or avocado) on the bed of rice in a manner that allows everything to be rolled together in the next step. Initial designs used a single column of fish, where a vertical conveyor belt system would drop one piece of fish as the platform positioned itself below. To simplify this further, we widened each compartment in the vertical column so that the machine just had to drop its floor in order to release the ingredients held inside the bottom compartment. This way, the platform could be positioned directly below the machine and the fish could be dropped onto it all at once in the desired position. With this vertical column, it acts as a queueing system for the ingredients and the upcoming rolls. For example, if the first sushi roll needs salmon and the second sushi roll needs tuna, the operator can simply put 4 pieces of salmon in the bottom compartment and 4 pieces of tuna in the second compartment. Our final iteration simply adds a duplicate vertical column side-by-side with the current column in order to allow for more variety in ingredients and better placements (ex. avocado should be placed in a row *next* to the fish, not on top of it).

After many design iterations for Station 3: Rolling the Sushi, we found the most success having the rolling mechanism built into the platform itself. Upon leaving Station 2, curled prongs hidden beneath the platform will reach up to grip the bamboo mat along with the sushi ingredients inside. By applying force with the stepper motor, the goal is for the ingredients to press in on itself and form a tight roll.

Station 4: Cutting the Roll uses a row of 5 blades in parallel that will come down to slice the roll into 6 pieces. We have decided to use straight blades that will come directly down onto the roll, but real-world tests will have to be performed to ensure that this is the best method of slicing. Nonetheless, safety is at the forefront in this station so clear plastic guards will be put around the blades so that humans do not have easy access. The guards will encapsulate the entirety of the sushi roll and retract when pressed down into the plate while slicing.



Preliminary design for Station 3: Rolling the Sushi

Finally, the platform will move to Station 5: Plating the Pieces. A UR3e robotic arm will be used as a pick-and-place robot, and the appropriate gripper end-effector will be attached. Ideally, the location of the sushi pickup and drop off points will be hardcoded into the robot. However, if this proves to be unfeasible during the execution stage, camera vision and processing will have to be considered.

Discussion

From the initial proposal phase of the project, the following timeline was put forward:

- *Weeks 1-3:* Brainstorm, assess, and finalize project ideas; create initial concepts
- *Week 4:* Present proposal
- *Weeks 5-6:* Narrow down design concepts; develop design prototypes; begin concept testing
- *Week 7:* Midterm report and presentation
- *Weeks 8-10:* Form BOMs, order key parts, build prototypes of key components
- *Week 11:* Create poster for presentation
- *Week 12:* Finalize work for the term; write final report
- *Week 13:* Poster presentations

Based on this, our group is solidly on track for the deliverables of Week 7 as we have finalized the designs for each station of our assembly line and started the initial prototypes for select parts. We have also started sourcing parts (ex. bamboo mat for rolling the sushi and wooden rollers for flattening the rice) and updating an ongoing bill of materials, as well as purchasing the key components of our system in order to begin concept testing.

By the start of the next term, we would like to have all our materials and planning completed so that we are ready to start building the project. This includes the finalized designs of each station, complete with specifications and measurements in a CAD file, an electrical diagram showing the IO wiring for each microcontroller, and a fully fleshed-out state diagram. The plan is to source and order parts over the summer to allow the vendors time to ship, or potentially custom-build, our desired components. This would give us the entirety of the fall semester to put the project together and face any issues that arise during this phase.

We are purposely allowing ourselves the most time to plan, test, and generate prototypes because we recognize the importance of creating high-quality designs from the start. The time investment put into the design stage now will ensure that we do not spend as much time reworking or redesigning the project during the execution stage next semester.

While the project seems feasible based on our current progress, we are prepared to narrow the scope as a contingency plan in the event that certain aspects of the sushi assembly line cannot

be achieved in time. For example, the initial proposal mentions using the HMI to select which ingredients would go into the roll. If this proves to be too difficult to implement into the program, we can scrap the idea without radically altering the design of the fish dispensing station. Additionally, a more drastic scope reduction would be to remove the entire automated section of flattening the rice. Instead, this would be a preliminary manual step for the chef to prepare beforehand.

Looking forward, a significant foreseeable issue stems from the ability to source the necessary parts in order to build each apparatus. With each complex, custom-designed machine, it may be difficult to find the right equipment that can meet all of our specifications as well as requirements (ie. it must be food-grade, be able to be disassembled for cleaning, etc.). This is why we are aiming to source these parts early so that we know what is available and what is not, and we can consider alternative options such as custom manufacturing the parts or using a different material.

With existing technologies like the AUTEC rice roller machine, we can look to their designs as inspiration for specific sections of our assembly line process. For example, it is evident that the process of flattening and outputting the rice can be accomplished with only two motors and two rolling pins. However, our project aims to provide a seamless, end-to-end automated solution for creating sushi, which has not been accomplished at this scale before.

If successful, this project could see use in high-end sushi restaurants to be used alongside professional chefs. The sushi assembly line is meant to take the burden of creating the simple, yet high demand rolls off the chefs, allowing them to focus on the restaurant's more complex and impressive dishes. It can also act as a smaller-scale prototype for the mass production of prepackaged basic sushi rolls for sale in grocery stores.

Conclusion

We will continue making progress on the project as the weeks go by. The main things to focus on are the mechanical designs, the CAD files, and shopping for parts for the designs and filling out our BOM.

References

[1] AUTEC, "Our Story," 2026. [Online]. Available: https://www.sushimachines.com/cd/about/. [Accessed 2 March 2026].

[2] FoodSafety, "Safe Selection and Handling of Fish and Shellfish," FoodSafety.gov, [Online]. Available: https://www.foodsafety.gov/blog/safe-selection-and-handling-fish-and-shellfish. [Accessed 2 3 2026].